

AULTBEA, United KingdomWMO: **030340**Lat: **57.8589N**Lon: **5.6328W**Elev: **10**StdP: **101.20**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.5	-0.2	-5.2	2.4	1.8	-3.9	2.7	2.0	16.9	7.2	15.5	7.9	1.9	100	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	21.0	16.1	19.3	15.2	18.0	14.4	17.0	20.0	15.9	18.3	15.1	17.3	4.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	11.2	18.4	14.7	10.5	17.2	14.0	10.0	16.1	47.9	20.0	44.6	18.3	42.2	17.4	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.6	11.1	DB	-3.2	24.8	1.3	2.1	-4.1	26.3	-4.9	27.5	-5.6	28.7	-6.5	30.2	(4)
(5)			WB	-3.8	19.0	1.3	2.0	-4.7	20.4	-5.5	21.6	-6.2	22.7	-7.1	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	5.4	5.3	6.2	8.0	10.0	12.3	14.1	14.1	12.7	10.2	7.6	5.9	(6)	
	DBStd	4.02	2.59	2.52	2.44	2.57	2.57	2.10	2.05	1.88	1.97	2.31	2.63	2.93	(7)	
	HDD10.0	732	143	132	120	70	31	3	0	0	2	26	78	128	(8)	
	HDD18.3	3283	401	364	377	311	258	181	132	133	168	252	322	384	(9)	
	CDD10.0	494	1	1	1	9	32	72	128	127	84	32	6	2	(10)	
	CDD18.3	3	0	0	0	0	0	0	2	2	0	0	0	0	(11)	
	CDH23.3	12	0	0	0	0	1	1	5	5	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	4.9	5.9	6.0	5.5	4.7	4.3	4.2	3.9	4.1	4.5	5.2	5.4	5.7	(14)	
(15) Precipitation	PrecAvg	1566	181	160	127	89	92	80	86	110	136	166	179	167	(15)	
	PrecMax	2175	311	379	335	133	216	142	149	257	240	258	278	279	(16)	
	PrecMin	1095	30	57	25	40	17	27	26	45	45	56	66	30	(17)	
	PrecStd	252	73	78	59	24	41	30	28	53	52	58	57	74	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	12.8	14.1	18.2	21.4	21.7	23.6	23.6	20.5	17.5	14.8	13.4	(19)	
		MCWB	9.5	10.0	9.6	12.0	14.3	16.6	17.7	17.5	15.2	13.5	11.7	11.0	(20)	
	2%	DB	10.8	11.0	11.5	15.0	18.1	19.1	20.9	20.3	18.3	15.6	13.0	12.0	(21)	
		MCWB	9.3	9.4	8.9	10.6	12.5	14.9	16.6	16.2	14.5	12.7	11.0	10.4	(22)	
	5%	DB	10.0	10.1	10.4	13.2	15.8	17.3	19.0	18.6	17.0	14.5	12.2	11.2	(23)	
		MCWB	8.8	8.7	8.5	9.7	11.7	13.8	15.6	15.1	13.9	12.1	10.5	9.8	(24)	
	10%	DB	9.2	9.2	9.6	11.9	14.2	15.7	17.5	17.2	15.8	13.5	11.4	10.4	(25)	
		MCWB	7.9	7.9	8.1	9.1	10.8	12.8	14.7	14.4	13.3	11.5	9.8	9.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.7	10.6	12.7	15.2	17.3	18.8	18.2	16.3	14.4	12.8	11.7	(27)	
		MCDB	11.4	11.5	12.5	16.1	20.4	21.0	22.2	22.2	18.5	16.1	14.1	12.5	(28)	
	2%	WB	9.9	9.8	9.7	11.4	13.4	15.4	17.3	16.7	15.2	13.4	11.6	10.8	(29)	
		MCDB	10.5	10.7	10.7	14.0	16.7	18.2	20.1	19.5	17.3	15.0	12.6	11.7	(30)	
	5%	WB	9.0	9.0	9.0	10.4	12.2	14.3	16.1	15.7	14.4	12.6	10.7	10.1	(31)	
		MCDB	9.9	9.9	10.1	12.4	15.0	16.6	18.2	18.0	16.3	14.1	11.8	11.1	(32)	
	10%	WB	8.1	8.0	8.3	9.5	11.2	13.3	15.1	14.7	13.7	11.8	9.9	9.1	(33)	
		MCDB	9.0	9.0	9.5	11.4	13.7	15.2	17.2	16.7	15.4	13.3	11.3	10.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.5	4.9	5.8	6.3	5.5	5.3	5.2	5.2	4.7	4.3	4.4	(35)	
		MCDBR	4.2	4.8	5.8	8.0	9.5	8.5	8.1	7.8	7.1	5.9	4.7	4.9	(36)	
	5% WB	MCWBR	4.1	4.0	4.2	4.9	5.6	5.2	4.7	4.5	4.5	4.2	4.2	4.8	(37)	
		MCWBR	4.1	4.1	4.9	6.7	8.4	7.5	7.5	7.1	6.0	5.0	4.6	4.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.341	0.352	0.369	0.384	0.385	0.376	0.380	0.369	0.363	0.347	0.328	0.323	(40)	
		taud	2.136	2.241	2.280	2.297	2.349	2.413	2.418	2.455	2.449	2.371	2.218	2.103	(41)	
	Edn at Noon		468	666	770	824	850	862	850	835	780	672	489	365	(42)	
		Edh at Noon	56	77	96	111	112	107	104	94	82	67	50	42	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.35	0.95	1.99	3.42	4.60	4.52	4.23	3.39	2.31	1.20	0.48	0.24	(44)	
	RadStd		0.05	0.12	0.16	0.24	0.34	0.43	0.37	0.21	0.16	0.12	0.05	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page